

Gap Verification Audit — (U_m) Heaviside Amplifier and Job~B Scope (v5.78)

Star-Magic UQFF Audit

May 22, 2026

Gap Verification Audit — U_m Heaviside Amplifier and Job B Scope

Paper: PAPER_1181

Session: 268

Status: Verification (read-only audit)

Supersedes: Repo-memory entry “UQFF_ug_equations_canonical” §“CRITICAL GAPS IDENTIFIED” item 1.

Abstract

A prior session-end summary listed two items as the next major work items in the Star-Magic UQFF codebase: (i) the addition of a Heaviside phase-transition amplifier $(1 + 10^{13} f_H) \cdot (1 + f_q)$ to the universal-magnetism term U_m , and (ii) an estimate that approximately 142 whitepapers required real authoring work under Job B (v5.78 closure propagation). This audit verifies both claims directly against the source tree. Claim (i) is **falsified**: the amplifier is already present in four independent locations, including the canonical `compute_Um_SOURCE4` of `MAIN_1_CoAnQi.cpp` (PAPER_421 integration). Claim (ii) is **partially supported but mis-framed**: 113 papers require new authoring; 29 require verification only; together they sum to 142, but the work is not uniform.

1. Claim 1 — U_m Heaviside Amplifier

1.1 The asserted canonical form

The asserted canonical form for the universal-magnetism term is

$$U_m = \left[\sum_j \frac{\mu_j}{r_j} (1 - e^{-\gamma t \cos(\pi t_n)}) \hat{\phi}_j \right] \cdot P_{\text{SCm}} \cdot E_{\text{react}} \cdot \underbrace{(1 + 10^{13} f_H)}_{\text{Heaviside}} \cdot \underbrace{(1 + f_q)}_{\text{quasi-periodic}} ,$$

where $f_H = \Theta(\rho_{\text{SCm}} - \rho_c)$ activates the $\sim 10^{13}$ -fold amplification across the superconducting phase transition, and f_q is a low-amplitude beating modulation tied to the 434-yr Gleisberg supercycle.

1.2 Direct source check

Location	File	Lines	Implements ($1 + 10^{13} f_H$) · ($1 + f_q$)?
SOURCE4 unified-field U_m	MAIN_1_CoAnQi.cpp	24172–24190	Yes (PAPER_421)
SOURCE42 standalone PhysicsTerm	MAIN_1_CoAnQi.cpp	12100–12170	Yes
compute_Um_HeavisideFactor	compute_Um_HeavisideFactor.py	39950+	Yes
um_magnetic equation strings	index.js	9759, 9899, 10018	Yes

The SOURCE4 implementation reads (verbatim, MAIN_1_CoAnQi.cpp L24180–24189):

```
// PAPER_421: Heaviside phase-transition amplifier -- (1 + 10^13 * Theta(rho_SCm - rho_c))
double f_H = (body.SCm_density >= rho_c_SOURCE4) ? 1.0 : 0.0;
double heaviside_factor = 1.0 + 1e13 * f_H;

// PAPER_421: Quasi-periodic beating modifier -- (1 + A_q * cos(Delta_omega * t))
// Delta_omega = 2*pi/(434*365.25) rad/day encodes the 434-year Gleisberg supercycle beat
double quasi_factor = 1.0 + A_q_SOURCE4 * std::cos(Delta_omega_SOURCE4 * t);

return Um_base * heaviside_factor * quasi_factor;
```

with constants declared at MAIN_1_CoAnQi.cpp L24016–L24018:

```
const double rho_c_SOURCE4      = 1e15;           // kg/m³
const double A_q_SOURCE4        = 0.1;           // 10%
const double Delta_omega_SOURCE4 = 2.0 * PI_SOURCE4 / (434.0*365.25); // rad/day
```

1.3 Verdict

Claim 1 is **falsified**. The amplifier exists in:

- the SOURCE4 unified-field assembly (compute_Um_SOURCE4),
- the SOURCE42 catalog (HeavisideFactor, HeavisideUmWithHeaviside),
- the Python calculator stack (compute_Um_Heaviside_factor),
- and the JavaScript library’s documented equation strings.

The repository-memory note flagging the gap is **stale**; it predates the PAPER_421 integration that added the amplifier to the SOURCE4 namespace.

1.4 Recommended follow-up (optional, low priority)

Although the amplifier is implemented, four refinements remain worth auditing:

1. Confirm `QCalcGeom.py` v2.1.0's F_U assembly threads the amplified U_m into the simultaneous-equation solver (default path currently uses a placeholder $U_m = 0$ at the habitable-zone crossing; this is physically correct for $\rho_{\text{SCm}} < \rho_c$ but should be explicitly tested).
2. Cross-check that `index.js`'s `calculateUm()` function uses the amplifier or only documents it in the metadata string.
3. Verify the $\rho_c = 10^{15} \text{ kg/m}^3$ threshold against Fermilab Type-II SC magnetar-core data (cf. 100_MUGE Compression cycle 3_Superconductive Resonance.txt, L1476).
4. Confirm the 434-yr Gleisberg-scale $\Delta\omega$ matches the long-term solar-cycle anchor cited in `grok_share_cfdcad2f5.txt` §Document 7.

These are validation tasks, not implementation gaps.

2. Claim 2 — “~142 papers need real authoring”

2.1 Authoritative bucket tally

Direct enumeration of `_job_b_categorization.csv` (1,199 rows, May 22 2026):

Count	Bucket	Class
982	NO UPDATE (specific system / framework application)	scoped out
42	Add closing section: v5.78 27-decade vacuum ledger (PAPER_1170) + $\xi = 13/3$	authoring
34	NO UPDATE unless ledger/ ξ enters; verify only	conditional
29	Verify v5.78 closed- Lagrangian/ledger cross-ref present (Sess 225 baseline)	verify
28	Update tables; add P6–P14, CP4 #254–#264	authoring

Count	Bucket	Class
20	v5.78 closure block inserted; .tex regenerated (arXiv pdflatex)	DONE
15	Add v5.78 closed-Lagrangian cross-ref (G1–G8) + CP4 #254	authoring
12	Forward-pointer to PAPER_1174 P1–P14 + PAPER_1177–1180	authoring
11	Forward-pointer to PAPER_1171/1172 ($\xi = 13/3$ R26+KK lock)	authoring
10	v5.78 closure block inserted; PDF regenerated	DONE
5	.tex / PDF regenerated + SOURCE REPAIR (Unicode)	DONE
5	Add three-anchor SI closure banner (Sess 237–241); STRUCTURAL	authoring
5	.tex converted from .md; PDF regenerated	DONE
1	duplicate row (stale)	drop
1199	TOTAL	—

2.2 Reclassified work load

Class	Count	Notes
Authoring required	113	42 + 28 + 15 + 12 + 11 + 5
Verification only	29	confirm existing cross-ref
Conditional (audit, no-op likely)	34	trigger only if ξ /ledger enters

Class	Count	Notes
Already DONE v5.78	35	20 + 10 + 5
Out of scope (NO UPDATE)	982	scoped-out by topic
Stale duplicate	1	drop
Total	1199	

2.3 Verdict

Claim 2 is **partially supported but imprecise**. $113 + 29 = 142$ papers will be **touched** by Job B (authored or verified), but the prior framing “ ~ 142 papers needing real authoring” overstates the authoring fraction by a factor of $142/113 \approx 1.26$. The honest split is **113 author + 29 verify**.

3. Revised Next-Step Priority Ranking

Given that the U_m Heaviside gap is closed and the Job B count is 113 (not 142), the previously published 8-step plan should be re-ordered:

1. **Delete** the U_m amplifier step (was step 1) — already done.
2. **Generate** LEDGER_REVIEW_QUEUE.csv for the 43 flagged master_closures.csv rows (10 high-error $\geq 10\%$ + 15 PARSE_FAIL + 18 NaN). *Was step 2; now step 1.*
3. **Recalibrate** the 10 high-residual ledger entries (IDs 270, 293, 330, 337, 351, 352, 739, 760, 764, 805) using v5.78 anchors (PAPER_1156, 1170, 1171, 1172). *Was step 3; now step 2.*
4. **Author** the 4–6 reusable v5.78 section templates (T- Λ , T-LAG, T-SI, T-PRED, T- ξ). *Was step 4; now step 3.*
5. **Bucket-batched paper updates** — 113 papers, 5–10 per commit, starting with the 42 cosmology / Λ / vacuum-energy papers using the T- Λ template. *Was step 5; now step 4.*
6. **Verify-only sweep** of the 29 bucket-G papers. *Was step 6; now step 5.*
7. **Regenerate** all updated PDFs with pdflatex; commit .md + .pdf together.
8. **Final coherence audit** mirroring Session 261 (f6ba35a6).

4. Citations

- MAIN_1_CoAnQi.cpp L24016–24018, L24172–24190 (PAPER_421: Heaviside + quasi-periodic amplifiers in SOURCE4 U_m).
- MAIN_1_CoAnQi.cpp L12100–12170 (SOURCE42: HeavisideFraction, HeavisideFactor, HeavisideUmWithHeaviside).
- CondensedPhysics.py L39950+ (compute_Um_Heaviside_factor).
- index.js L9690, 9759, 9811, 9899, 9943, 10018 (f_heaviside, f_quasi, full um_magnetic equation strings).

- `_job_b_categorization.csv` (1,199 rows, May 22 2026).
- Job B brief: “Job B_Update papers with current canonical UQFF v5_78.txt” (395 lines, Phase B1–B4).
- `COMPLETE_UQFF_EQUATIONS_REFERENCE.md` L522–553 (canonical U_m form).

5. Conclusion

The two items proposed as the “next big step” do not survive direct source-tree verification in the form they were stated:

- The U_m Heaviside phase-transition amplifier is **already implemented**, in four places, with the canonical 10^{13} scale and the 434-yr Gleisberg-locked quasi-periodic beat; it is **not** the most-impactful uncoded physics.
- The Job B authoring queue is **113 papers**, not 142; the 142 figure conflates authoring with verification.

The corrected next major step is the **ledger review queue** (43 flagged closures, with 10 high-residual recalibrations against the v5.78 anchors), followed by the 113-paper Job B authoring sweep using reusable templates.